

SkyTech Troubleshooting Guide

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How to find and fix any problem in the SkyTech Admin site, database, scripts and GitHub releases. Every error shows a SkyTech code; look it up here. Start with First aid if you are not sure where the problem is.

Contents

- 1. [First aid in five steps](#)
- 2. [How logging works](#)
- 3. [Code index](#)
- 4. [All error codes](#)
- 5. [Predicted errors \(prevention\)](#)
- 6. [SQL Server error numbers](#)
- 7. [Database error and audit queries](#)
- 8. [Manual recovery recipes](#)

1. First aid in five steps

Step	What to do
1. Read the code	Every SkyTech message ends with a code such as SKY-DB-001 and often a reference (ref 3f9c2a1b7d04). Find the code below — each card says what it means, the likely causes and the fix steps in order.
2. Run the doctor	Open a Command Prompt in admin-site and run npm run doctor . It checks Node.js, packages, .env and secrets, network exposure, git, log folder, disk, port and the database (objects, unique rules, logins, recent database errors, app role, last backup), and prints each problem with its code and fix. Use npm run doctor -- --demo without a database.
3. Look in the log	Admin site → System log (admins), or open logs\runtime\admin-YYYY-MM-DD.log in Notepad and search for the ref value or the code. Database procedures write to dbo.ErrorLog (queries below). Scripts write logs\runtime\scripts-YYYY-MM-DD.log.
4. Fix and re-test	Apply the first fix step, repeat the action, and run the doctor again. Most fixes are settings in admin-site\.env, re-running a SQL script in SSMS (all are safe to re-run), or restarting the Admin site.

Step	What to do
5. Still stuck?	Send SkyTech_Manager the code, the ref value and the doctor output (never send passwords or the .env file).

2. How logging works

Part	Where	What is recorded
Admin site	logs\runtime\admin-YYYY-MM-DD.log	One JSON line per event: ts, level, category (http, auth, security, db, demo, startup), msg, code, reqId, user, ip, status, ms. Every API call, sign-in, logout, CSRF/rate-limit block, permission refusal and error is recorded.
Doctor	logs\runtime\doctor-YYYY-MM-DD.log	Each check and its result, with the code.
Scripts (Node + Python)	logs\runtime\scripts-YYYY-MM-DD.log	build-demos, lead builders, document builders: start, files written, failures with code and (for crashes) the traceback.
Database	SkyTechCRM.dbo.ErrorLog	One row per failure caught inside a SkyTech procedure: time, procedure, SQL error number, line, message, SkyTech code, context, login, computer.
Database audit	SkyTechCRM.dbo.AuditLog	Who signed in, created, changed, merged or deleted what (append-only; protected by trigger, SKY-SEC-006).

- Levels: debug < info < warn < error. LOG_LEVEL in .env picks the minimum written (default info).
- Passwords, hashes, secrets, tokens, cookies and CSRF values are replaced by [redacted] before anything is written.
- Files older than LOG_RETENTION_DAYS (default 30) are deleted automatically; the logs\runtime folder is git-ignored and never uploaded.
- The ref shown on screen (X-Request-Id) is the reqId in the log — search for it to see exactly what happened.
- If the log folder cannot be written the site keeps running and logs to the screen only (SKY-SYS-003).

3. Code index

67 codes in 13 areas. Format SKY-<AREA>-<NNN>; severity: critical (stops the site or risks data), error (an action failed), warn (attention needed), info (normal refusal, e.g. wrong password). Source: config/error-codes.json V1.3 — the same file the Admin site uses, so screen, log and guide always match.

Area	Codes
CFG — Configuration	SKY-CFG-001 SKY-CFG-002 SKY-CFG-003 SKY-CFG-004 SKY-CFG-005 SKY-CFG-006 SKY-CFG-007
DB — Database	SKY-DB-001 SKY-DB-002 SKY-DB-003 SKY-DB-004 SKY-DB-005 SKY-DB-006 SKY-DB-007 SKY-DB-008 SKY-DB-009 SKY-DB-010
AUTH — Sign-in	SKY-AUTH-001 SKY-AUTH-002 SKY-AUTH-003 SKY-AUTH-004 SKY-AUTH-005 SKY-AUTH-006 SKY-AUTH-007 SKY-AUTH-008
SEC — Security	SKY-SEC-001 SKY-SEC-002 SKY-SEC-003 SKY-SEC-004 SKY-SEC-005 SKY-SEC-006 SKY-SEC-007
DATA — Data	SKY-DATA-001 SKY-DATA-002 SKY-DATA-003 SKY-DATA-004 SKY-DATA-005 SKY-DATA-006
DUP — Duplicates	SKY-DUP-001 SKY-DUP-002 SKY-DUP-003 SKY-DUP-004 SKY-DUP-005
DEMO — Demo sites	SKY-DEMO-001 SKY-DEMO-002 SKY-DEMO-003 SKY-DEMO-004 SKY-DEMO-005
IMP — Imports	SKY-IMP-001 SKY-IMP-002 SKY-IMP-003
LEAD — Lead scripts	SKY-LEAD-001 SKY-LEAD-002 SKY-LEAD-003
DOC — Documents	SKY-DOC-001 SKY-DOC-002
GIT — GitHub	SKY-GIT-001 SKY-GIT-002 SKY-GIT-003 SKY-GIT-004 SKY-GIT-005
BAK — Backup	SKY-BAK-001 SKY-BAK-002 SKY-BAK-003
SYS — System	SKY-SYS-001 SKY-SYS-002 SKY-SYS-003

4. All error codes

CFG — Configuration

SKY-CFG-001 — Settings file .env is missing critical

Where: Admin site start (npm start / start-admin.bat) · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number); screen

Message shown: Settings file .env is missing

Likely causes

- The admin-site\.env file was never created
- It was renamed or deleted

Fix (in this order)

1. In the admin-site folder copy .env.example to a new file named .env
2. Fill in DB_SERVER, DB_AUTH, DB_DRIVER and SESSION_SECRET (see admin-site README)
3. Run npm run doctor to confirm, then start the site again

SKY-CFG-002 — SESSION_SECRET is missing or too weak critical

Where: Admin site start · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number); screen

Message shown: SESSION_SECRET is missing or too weak

Likely causes

- SESSION_SECRET is empty, still 'change-this...', or shorter than 32 characters

Fix (in this order)

1. Open admin-site\.env
2. Set SESSION_SECRET to a random phrase of at least 32 characters (npm run doctor -- --new-secret prints one)
3. Save and start the site again; everyone must sign in again

SKY-CFG-003 — Site opened to the network without HTTPS cookies critical

Where: Admin site start · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number); screen

Message shown: Site opened to the network without HTTPS cookies

Likely causes

- HOST is not 127.0.0.1/localhost but COOKIE_SECURE is not 1

Fix (in this order)

1. Keep HOST=127.0.0.1 for local use (recommended)
2. Only when hosting behind HTTPS: set COOKIE_SECURE=1 and TRUST_PROXY=1

SKY-CFG-004 — Node.js version is too old error

Where: Admin site start, doctor · **Logged in:** logs/runtime/doctor-YYYY-MM-DD.log and the doctor screen

Message shown: Node.js version is too old

Likely causes

- Node.js older than version 18 is installed

Fix (in this order)

1. Download the LTS version from <https://nodejs.org> and install it
2. Close and reopen the command window, then run node --version

SKY-CFG-005 — demo-sites folder not found warn

Where: Admin site: Save to demo-sites folder · HTTP 500 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: demo-sites folder not found

Likely causes

- admin-site was moved away from the repository folder
- DEMO_SITES_DIR points to a folder that does not exist

Fix (in this order)

1. Keep admin-site inside skytech-agent-program next to demo-sites
2. Or set DEMO_SITES_DIR in .env to the full path of the demo-sites folder

SKY-CFG-006 — Port already in use error predicted

Where: Admin site start · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number); screen

Message shown: Port already in use

Likely causes

- The Admin site is already running in another window
- Another program uses port 3030

Fix (in this order)

1. Close the other black Admin site window, or
2. Set PORT=3031 in .env and open http://127.0.0.1:3031

SKY-CFG-007 — Required package missing error

Where: Admin site start, doctor, build-demos · **Logged in:** logs/runtime/doctor-YYYY-MM-DD.log and the doctor screen

Message shown: Required package missing

Likely causes

- npm install was not run, or node_modules was deleted

Fix (in this order)

1. Open a Command Prompt in admin-site and run npm install
2. Run npm run doctor

DB — Database

SKY-DB-001 — Cannot connect to SQL Server critical

Where: Admin site start, create-admin, doctor · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number); logs/runtime/doctor-YYYY-MM-DD.log and the doctor screen

Message shown: Cannot connect to SQL Server

Likely causes

- The SQL Server service is stopped
- DB_SERVER does not match the server name used in SSMS
- Firewall or instance name wrong (e.g. localhost\SQL2014)

Fix (in this order)

1. Press Windows key → Services → 'SQL Server (...)' → Start
2. Open SSMS, copy the exact Server name into DB_SERVER in .env
3. Run npm run doctor; if it still fails, try npm run demo to confirm the site itself works

SKY-DB-002 — Windows login needs the msnodesqlv8 package critical

Where: Admin site start (DB_AUTH=windows) · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: Windows login needs the msnodesqlv8 package

Likely causes

- msnodesqlv8 failed to install (it only installs on Windows)

Fix (in this order)

1. On the Windows PC run npm install in admin-site
2. If it still fails, install 'Visual C++ Redistributable', or use DB_AUTH=sql with a SQL login (see sql/06)

SKY-DB-003 — ODBC driver not found critical

Where: Admin site start (DB_AUTH=windows) · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: ODBC driver not found

Likely causes

- DB_DRIVER names a driver that is not installed

Fix (in this order)

1. Windows key → 'ODBC Data Sources (64-bit)' → Drivers tab
2. Copy an installed name (e.g. 'SQL Server Native Client 11.0' or 'ODBC Driver 17 for SQL Server') into DB_DRIVER

SKY-DB-004 — Database SkyTechCRM does not exist critical

Where: Admin site start, doctor · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number); logs/runtime/doctor-YYYY-MM-DD.log and the doctor screen

Message shown: Database SkyTechCRM does not exist

Likely causes

- sql/01_create_skytechcrm.sql has not been run
- DB_DATABASE has a different name

Fix (in this order)

1. In SSMS open and run sql/01_create_skytechcrm.sql, then the sample import, sql/04, sql/05 and sql/06
2. Check DB_DATABASE=SkyTechCRM in .env

SKY-DB-005 — A database table, view or procedure is missing error

Where: Any Admin site screen · HTTP 500 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number); logs/runtime/doctor-YYYY-MM-DD.log and the doctor screen

Message shown: A database table, view or procedure is missing

Likely causes

- A newer script version was not run in SSMS (e.g. sql/04 or sql/06)

Fix (in this order)

1. Run npm run doctor to see which object is missing
2. Run the SQL scripts again in order: 01 → sample import → 04 → 05 → 06 (all are safe to re-run)

SKY-DB-006 — SQL login failed critical

Where: Admin site start (DB_AUTH=sql) · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: SQL login failed

Likely causes

- DB_USER or DB_PASSWORD wrong
- SQL Server authentication (mixed mode) is turned off
- The login is disabled or its password expired

Fix (in this order)

1. Check DB_USER / DB_PASSWORD in .env
2. In SSMS: server Properties → Security → 'SQL Server and Windows Authentication mode', restart the service
3. Or switch to DB_AUTH=windows

SKY-DB-007 — Database request timed out error predicted

Where: Any Admin site screen · HTTP 503 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: Database request timed out

Likely causes

- SQL Server is busy or a long query is blocking
- Very large table without index

Fix (in this order)

1. Wait and retry
2. In SSMS run: EXEC sp_who2 to look for blocking sessions
3. Restart the SQL Server service if it stays stuck

SKY-DB-008 — Permission denied in the database error predicted

Where: Any Admin site screen · HTTP 500 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: Permission denied in the database

Likely causes

- The app connects with a restricted login that lacks a permission
- sql/06 role was created before a new table existed

Fix (in this order)

1. Re-run sql/06_security_and_logging.sql (re-grants all permissions to SkyTechApp)
2. Make sure the app's login is a member of the SkyTechApp role

SKY-DB-009 — Unexpected database error error

Where: Any database action · HTTP 500 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number); dbo.ErrorLog in SkyTechCRM (SELECT TOP 50 * FROM dbo.ErrorLog ORDER BY ErrorLogID DESC)

Message shown: Unexpected database error

Likely causes

- An error not covered by other codes

Fix (in this order)

1. Note the reference number shown on screen
2. Search the admin log for that reference to read the SQL error number and message
3. Look the SQL error number up in this guide or send it to SkyTech_Manager

SKY-DB-010 — Database errors were recorded in dbo.ErrorLog warn predicted

Where: npm run doctor; any stored procedure · **Logged in:** dbo.ErrorLog (database)

Likely causes

- A stored procedure failed (import, merge, fix, backup) and wrote its error to dbo.ErrorLog before rolling back

Fix (in this order)

1. In SSMS run: SELECT TOP 20 * FROM SkyTechCRM.dbo.ErrorLog ORDER BY ErrorLogID DESC
2. Look up the ErrorNumber in the Troubleshooting Guide table "SQL Server error numbers"
3. After fixing, old rows can be removed with EXEC dbo.usp_PurgeErrorLog @KeepDays = 90

AUTH — Sign-in

SKY-AUTH-001 — Wrong username or password info

Where: Admin site sign-in · HTTP 401 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: Wrong username or password

Likely causes

- Typing error
- Account disabled (IsActive = 0)

Fix (in this order)

1. Check Caps Lock and try again
2. Reset with npm run create-admin using the same username

SKY-AUTH-002 — Too many sign-in attempts — locked for 15 minutes warn

Where: Admin site sign-in · HTTP 429 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: Too many sign-in attempts — locked for 15 minutes

Likely causes

- 5 wrong passwords for this username or from this computer within 15 minutes

Fix (in this order)

1. Wait 15 minutes, or restart the Admin site (clears the lock)
2. If you did not cause it, check the security log for LOGIN_FAIL entries

SKY-AUTH-003 — Session expired info

Where: Any Admin site screen · HTTP 401 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: Session expired

Likely causes

- No activity for 30 minutes (SESSION_IDLE_MIN), or signed in longer than 8 hours

Fix (in this order)

1. Sign in again

SKY-AUTH-004 — Not signed in info

Where: Any Admin site request · HTTP 401 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: Not signed in

Likely causes

- The browser has no session (new tab, cookies cleared, site restarted)

Fix (in this order)

1. Sign in again

SKY-AUTH-005 — Action not allowed for your role warn

Where: Create, edit, delete, duplicates, demo save, Logins, System log · HTTP 403 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: Your role cannot do this. Ask an admin to change your role on the Logins screen.

Likely causes

- Your login's role does not include this action (roles: admin, manager, sales, webdev, viewer — see admin-site/src/roles.js)
- A viewer tried to change data

Fix (in this order)

1. Ask an admin to change your role on the Logins screen
2. Or sign in with a login that has the right role

SKY-AUTH-006 — Password does not meet the policy warn

Where: npm run create-admin; Admin site → Logins (new login, reset password) and Change my password · HTTP 400 · **Logged in:** logs/runtime/scripts-YYYY-MM-DD.log and the script's screen output

Message shown: Password does not meet the policy

Likely causes

- Fewer than 12 characters, missing upper/lower case or a digit, contains the username, or is a common password

Fix (in this order)

1. Choose a passphrase of 12+ characters with upper and lower case letters and a number

SKY-AUTH-007 — Your password is needed to confirm this change warn

Where: Admin site → Logins; Change my password · HTTP 403 · **Logged in:** admin log (category security); dbo.AuditLog

Message shown: Your password is needed to confirm this change (it was empty or wrong).

Likely causes

- The "Your password" box was empty or wrong when creating, changing or resetting a login
- Too many wrong confirmation attempts also lock the account for 15 minutes (SKY-AUTH-002)

Fix (in this order)

1. Type your own current password in the "Your password" box and save again
2. Forgotten it? Another admin can reset it on the Logins screen, or run npm run create-admin with your username

SKY-AUTH-008 — Change refused: it would lock everyone out warn

Where: Admin site → Logins (edit role or active) · HTTP 409 · **Logged in:** admin log (category http)

Message shown: Not allowed: there must always be at least one active admin, and you cannot disable or demote yourself.

Likely causes

- You tried to disable your own login or remove your own admin role
- The change would leave no active admin login

Fix (in this order)

1. Make another login an active admin first, then change this one
2. To lock out a person, disable their login — not your own

SEC — Security

SKY-SEC-001 — Security token missing or invalid (CSRF) warn

Where: Any change in the Admin site · HTTP 403 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: Security token missing or invalid (CSRF)

Likely causes

- The page was open before the site restarted
- Another website tried to send a request on your behalf

Fix (in this order)

1. Reload the page (F5) and repeat the action
2. If it happens without reason, check the security log

SKY-SEC-002 — Too many requests — slow down warn

Where: Admin site · HTTP 429 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: Too many requests — slow down

Likely causes

- More than the allowed number of requests per minute from this computer (a stuck script or an attack)

Fix (in this order)

1. Wait one minute and retry
2. Close duplicate browser tabs running auto-refresh

SKY-SEC-003 — Request too large warn

Where: Admin site save · HTTP 413 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: Request too large

Likely causes

- A field or upload exceeded 200 KB

Fix (in this order)

1. Shorten the text and save again

SKY-SEC-004 — Possible secret found in files to be uploaded critical

Where: push-to-github.bat (secret scan) · **Logged in:** Screen output of the secret scan

Message shown: Possible secret found in files to be uploaded

Likely causes

- A password, token, private key or connection string with a password is in a tracked file

Fix (in this order)

1. Open the file and line shown by the scanner
2. Move the secret into admin-site\.env (never committed) and remove it from the file
3. Commit the fix, then run push-to-github.bat again; if a real secret was already pushed, change that password/token

SKY-SEC-005 — Photo link not allowed warn

Where: Admin site → Demo sites · HTTP 400 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: Photo link not allowed

Likely causes

- A photo URL is not from images.pexels.com or images.unsplash.com

Fix (in this order)

1. Use a free-licence photo link from Pexels or Unsplash

SKY-SEC-006 — Audit log is protected and cannot be changed critical predicted

Where: Database (dbo.AuditLog) · **Logged in:** dbo.ErrorLog in SkyTechCRM (SELECT TOP 50 * FROM dbo.ErrorLog ORDER BY ErrorLogID DESC)

Message shown: Audit log is protected and cannot be changed

Likely causes

- Someone tried to UPDATE or DELETE audit records

Fix (in this order)

1. Audit records are append-only by design; nothing to fix
2. Investigate who tried, using SQL Server logs

SKY-SEC-007 — Request from another website blocked warn

Where: Admin site · HTTP 403 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: Request from another website blocked

Likely causes

- The request's Origin header does not match the Admin site address

Fix (in this order)

1. Use the Admin site only from its own address (http://127.0.0.1:3030)

DATA — Data

SKY-DATA-001 — Required field missing info

Where: Admin site save · HTTP 400 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: Required field missing

Likely causes

- A field marked * was left empty

Fix (in this order)

1. Fill in the fields marked * and save again

SKY-DATA-002 — Value has the wrong format info

Where: Admin site save · HTTP 400 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: Value has the wrong format

Likely causes

- Colour not like #1f5fbf, slug with spaces/capitals, bad date or number, or the request was not valid JSON

Fix (in this order)

1. Correct the field shown in the message and save again

SKY-DATA-003 — Record is linked to other records warn

Where: Admin site delete · HTTP 409 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: Record is linked to other records

Likely causes

- Deleting a company that still has a lead, web presence or demo site, or a lead with activities

Fix (in this order)

1. Delete or move the linked records first, or use Possible duplicates → merge

SKY-DATA-004 — Record not found info

Where: Admin site · HTTP 404 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: Record not found

Likely causes

- It was deleted or merged in another tab

Fix (in this order)

1. Reload the list

SKY-DATA-005 — This view is read-only info

Where: Admin site · HTTP 405 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: This view is read-only

Likely causes

- Trying to change Imports, Possible duplicates or Audit log directly

Fix (in this order)

1. Change the underlying record instead

SKY-DATA-006 — Unknown table warn

Where: Admin site API · HTTP 404 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: Unknown table

Likely causes

- A link or bookmark names a table the site does not allow

Fix (in this order)

1. Use the menu

DUP — Duplicates

SKY-DUP-001 — Duplicate record refused info

Where: Admin site save, imports · HTTP 409 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: Duplicate record refused

Likely causes

- Same source ID, same name + ZIP, a second lead/web row/demo site for one company

Fix (in this order)

1. Open Possible duplicates and merge, or edit the existing record instead

SKY-DUP-002 — Leads cannot be merged warn

Where: Possible duplicates → merge leads · HTTP 409 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number); dbo.ErrorLog in SkyTechCRM (SELECT TOP 50 * FROM dbo.ErrorLog ORDER BY ErrorLogID DESC)

Message shown: Leads cannot be merged

Likely causes

- The two leads no longer belong to the same company, or one was deleted (SQL 50011/50012)

Fix (in this order)

1. Click Check now to refresh the list
2. Merge the companies first if needed

SKY-DUP-003 — Nothing to fix — company has one web-presence row info

Where: Possible duplicates → fix web presence · HTTP 409 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: Nothing to fix — company has one web-presence row

Likely causes

- Already fixed in another tab (SQL 50021)

Fix (in this order)

1. Click Check now

SKY-DUP-004 — Unique rule could not be created warn predicted

Where: sql/01 or sql/04 run in SSMS · **Logged in:** SSMS Messages tab

Message shown: Unique rule could not be created

Likely causes

- Duplicates existed when the script ran, so the unique index was skipped (WARNING in Messages)

Fix (in this order)

1. Open Admin site → Possible duplicates, fix all Exact duplicates
2. Run the script again; the index is then created

SKY-DUP-005 — Companies cannot be merged warn

Where: Possible duplicates → merge companies (dbo.usp_MergeCompanies) · HTTP 409 · **Logged in:** admin log (category http/db); dbo.ErrorLog

Message shown: These two companies cannot be merged. Reload the list and try again.

Likely causes

- The same company was chosen twice (SQL error 50001)
- One of the two companies was deleted or merged by someone else (SQL error 50002)

Fix (in this order)

1. Reload the Possible duplicates screen and choose two different companies that still exist

DEMO — Demo sites

SKY-DEMO-001 — Design file is not valid error

Where: npm run build-demos · **Logged in:** logs/runtime/scripts-YYYY-MM-DD.log and the script's screen output

Message shown: Design file is not valid

Likely causes

- A comma or quote is missing in demo-sites\designs\<name>.json

Fix (in this order)

1. Open the file named in the message; check the line shown
2. Paste it into a JSON checker (e.g. jsonlint.com) if unsure, fix and rebuild

SKY-DEMO-002 — Design has no matching lead error

Where: npm run build-demos · **Logged in:** logs/runtime/scripts-YYYY-MM-DD.log and the script's screen output

Message shown: Design has no matching lead

Likely causes

- The OvertureID in the design file is not in data/samples/SkyTech_Leads_Sample100.csv

Fix (in this order)

1. Copy the correct OvertureID from the lead CSV into the design file

SKY-DEMO-003 — Unknown demo name error

Where: npm run build-demos -- <slug> · **Logged in:** logs/runtime/scripts-YYYY-MM-DD.log and the script's screen output

Message shown: Unknown demo name

Likely causes

- The slug typed does not match any design file

Fix (in this order)

1. Use the file name without .json, e.g. c-and-a-plumbing-20723

SKY-DEMO-004 — Photos do not appear on a demo warn predicted

Where: Demo page in a browser · **Logged in:** Browser developer tools (F12 → Console)

Message shown: Photos do not appear on a demo

Likely causes

- No internet connection
- Pexels changed or removed the photo
- A firewall blocks images.pexels.com

Fix (in this order)

1. Open the page online; try another photo link in Admin site → Demo sites
2. Before going live, download the photos and host them with the site (img/ folder)

SKY-DEMO-005 — Fonts look plain info predicted

Where: Demo page offline · **Logged in:** —

Message shown: Fonts look plain

Likely causes

- Google Fonts cannot load offline; a similar system font is used

Fix (in this order)

1. No action needed; fonts load when online

IMP — Imports

SKY-IMP-001 — Some demos were not loaded into the database warn

Where: sql/05 in SSMS · **Logged in:** SSMS results grid

Message shown: Some demos were not loaded into the database

Likely causes

- The result grid 'NotFoundInDatabase' lists businesses whose company is not in SkyTechCRM yet

Fix (in this order)

1. Run the lead import first (data/samples/..._import.sql), then sql/05 again

SKY-IMP-002 — Lead import failed and was rolled back error

Where: usp_ImportStagedLeads (lead import SQL) · **Logged in:** dbo.ErrorLog in SkyTechCRM (SELECT TOP 50 * FROM dbo.ErrorLog ORDER BY ErrorLogID DESC)

Message shown: Lead import failed and was rolled back

Likely causes

- Bad data in the staging rows (too long, invalid characters) or a missing object

Fix (in this order)

1. Read the newest row in dbo.ErrorLog for the exact SQL message
2. Fix the data or run sql/01 again, then re-run the import (nothing was half-imported)

SKY-IMP-003 — Staging table not empty warn predicted

Where: Lead import · **Logged in:** dbo.ErrorLog in SkyTechCRM (SELECT TOP 50 * FROM dbo.ErrorLog ORDER BY ErrorLogID DESC)

Message shown: Staging table not empty

Likely causes

- A previous import stopped half-way; dbo.StgLeads still has rows

Fix (in this order)

1. Nothing to do: each import clears staging at the start
2. If in doubt: DELETE FROM dbo.StgLeads; then re-run the import

LEAD — Lead scripts

SKY-LEAD-001 — Lead input file missing or unreadable error

Where: scripts/leads/*.py · **Logged in:** logs/runtime/scripts-YYYY-MM-DD.log and the script's screen output

Message shown: Lead input file missing or unreadable

Likely causes

- Path typed wrong, or the file is open in Excel

Fix (in this order)

1. Check the path in the command
2. Close the file in Excel and retry

SKY-LEAD-002 — Overture extract is empty error predicted

Where: scripts/leads (browser extract) · **Logged in:** logs/runtime/scripts-YYYY-MM-DD.log and the script's screen output

Message shown: Overture extract is empty

Likely causes

- Map tiles were not loaded when the extract ran

Fix (in this order)

1. Zoom the Overture Explorer map to the area, wait for points to appear, then run the extract again

SKY-LEAD-003 — Data source blocked by the network warn predicted

Where: Lead pulls · **Logged in:** logs/runtime/scripts-YYYY-MM-DD.log and the script's screen output

Message shown: Data source blocked by the network

Likely causes

- The Overture download bucket or OpenStreetMap API is blocked by the network policy

Fix (in this order)

1. Use the Overture Explorer in the Claude desktop browser (works)
2. Or download files manually and place them in data/raw/

DOC — Documents

SKY-DOC-001 — Document builder needs a missing tool error

Where: scripts/docs/*.py · **Logged in:** logs/runtime/scripts-YYYY-MM-DD.log and the script's screen output

Message shown: Document builder needs a missing tool

Likely causes

- Python package missing (e.g. reportlab for the prompt book)

Fix (in this order)

1. Run: pip install reportlab
2. PDF versions of documents are made by SkyTech_Manager; HTML/Markdown build without extra tools

SKY-DOC-002 — Document source could not be read error

Where: scripts/docs/*.py · **Logged in:** logs/runtime/scripts-YYYY-MM-DD.log and the script's screen output

Message shown: Document source could not be read

Likely causes

- A source file (error catalog, diagram data, code file) was moved or has a syntax error

Fix (in this order)

1. Run the builder again and read the file name in the message
2. Undo the last edit to that file (GitHub Desktop → History) and rebuild

GIT — GitHub

SKY-GIT-001 — GitHub rejected the sign-in error

Where: push-to-github.bat · **Logged in:** Screen output of push-to-github.bat or GitHub Desktop

Message shown: GitHub rejected the sign-in

Likely causes

- Saved GitHub password/token expired ('Invalid username or token')

Fix (in this order)

1. Windows key → Credential Manager → Windows Credentials → remove entries starting git:https://github.com
2. Run push-to-github.bat again and sign in in the browser window

SKY-GIT-002 — Git lock file left behind error

Where: push-to-github.bat, GitHub Desktop · **Logged in:** Screen output of push-to-github.bat or GitHub Desktop; logs/runtime/doctor-YYYY-MM-DD.log and the doctor screen

Message shown: Git lock file left behind

Likely causes

- A git command was interrupted ('index.lock' or 'HEAD.lock' ... File exists)

Fix (in this order)

1. Make sure GitHub Desktop is closed
2. Delete skytech-agent-program\.git\index.lock (or HEAD.lock) in File Explorer
3. Run the push again; npm run doctor reports leftover locks

SKY-GIT-003 — GitHub remote not set error

Where: push-to-github.bat · **Logged in:** Screen output of push-to-github.bat or GitHub Desktop

Message shown: GitHub remote not set

Likely causes

- The folder was copied without its GitHub link

Fix (in this order)

1. In this folder run: git remote add origin https://github.com/skytechsolutions2020/skytech-agent-program.git
2. Then run push-to-github.bat

SKY-GIT-004 — GitHub has changes this computer does not warn predicted

Where: push-to-github.bat · **Logged in:** Screen output of push-to-github.bat or GitHub Desktop

Message shown: GitHub has changes this computer does not

Likely causes

- Files were edited directly on github.com ('rejected ... fetch first')

Fix (in this order)

1. Open GitHub Desktop → Fetch origin → Pull, then push again
2. Do not edit files on github.com; edit in the local folder

SKY-GIT-005 — Git is not installed or this is not the repository folder error predicted

Where: push-to-github.bat, scan-secrets, doctor · **Logged in:** screen

Likely causes

- Git for Windows is not installed or not on PATH
- The script was run outside the skytech-agent-program folder

Fix (in this order)

1. Install Git for Windows (free) and reopen the window
2. Run the script from C:\Users\AV\Documents\SkyTechClaude\skytech-agent-program

BAK — Backup

SKY-BAK-001 — Database backup failed error

Where: usp_BackupSkyTechCRM (sql/06) · **Logged in:** dbo.ErrorLog in SkyTechCRM (SELECT TOP 50 * FROM dbo.ErrorLog ORDER BY ErrorLogID DESC)

Message shown: Database backup failed

Likely causes

- Backup folder does not exist or SQL Server's service account cannot write to it
- Disk full

Fix (in this order)

1. Use a folder SQL Server can write to (default C:\SkyTechBackups, created by the script) or grant the service account access
2. Free disk space and run EXEC dbo.usp_BackupSkyTechCRM again

SKY-BAK-002 — Database damaged or lost — restore needed critical predicted

Where: SQL Server · **Logged in:** dbo.ErrorLog in SkyTechCRM (SELECT TOP 50 * FROM dbo.ErrorLog ORDER BY ErrorLogID DESC)

Message shown: Database damaged or lost — restore needed

Likely causes

- Disk failure, accidental drop, corrupted database

Fix (in this order)

1. Find the newest SkyTechCRM_*.bak in the backup folder
2. In SSMS: right-click Databases → Restore Database → Device → choose the .bak → OK
3. Run npm run doctor to confirm

SKY-BAK-003 — No recent database backup warn predicted

Where: npm run doctor · **Logged in:** doctor log; msdb.dbo.backupset

Likely causes

- dbo.usp_BackupSkyTechCRM has not been run in the last 7 days (or never)

Fix (in this order)

1. In SSMS run: EXEC SkyTechCRM.dbo.usp_BackupSkyTechCRM;
2. Copy C:\SkyTechBackups to a USB drive or cloud folder once a week

SYS — System

SKY-SYS-001 — Unexpected error error

Where: Anywhere · HTTP 500 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: Unexpected error

Likely causes

- A situation the program did not expect

Fix (in this order)

1. Note the reference number on screen
2. Open the newest admin log and search for that reference; the code and details are there
3. Send the log lines to SkyTech_Manager

SKY-SYS-002 — Disk full critical predicted

Where: Logs, demo files, backups · HTTP 507 · **Logged in:** logs/runtime/admin-YYYY-MM-DD.log (search for the code or the reference number)

Message shown: Disk full

Likely causes

- The drive has no free space

Fix (in this order)

1. Free disk space (empty Recycle Bin, delete old backups beyond the last 5)
2. Restart the Admin site

SKY-SYS-003 — Log folder cannot be written warn

Where: Admin site, scripts · **Logged in:** Screen

Message shown: Log folder cannot be written

Likely causes

- logs/runtime is read-only or blocked by antivirus

Fix (in this order)

1. Make sure logs/runtime exists and is not read-only
2. The site keeps running and logs to the screen only

5. Predicted errors (prevention)

These have not happened yet but are expected as the program grows (more leads, a second PC, hosting on Hostinger, longer running time). Each has a prevention built in and a manual fix.

Code	Problem	Prevention already in place / check
SKY-CFG-006	Port already in use	Close the other black Admin site window, or
SKY-DB-007	Database request timed out	Wait and retry
SKY-DB-008	Permission denied in the database	Re-run sql/06_security_and_logging.sql (re-grants all permissions to SkyTechApp)
SKY-DB-010	Database errors were recorded in dbo.ErrorLog	In SSMS run: SELECT TOP 20 * FROM SkyTechCRM.dbo.ErrorLog ORDER BY ErrorLogID DESC
SKY-SEC-006	Audit log is protected and cannot be changed	Audit records are append-only by design; nothing to fix
SKY-DUP-004	Unique rule could not be created	Open Admin site → Possible duplicates, fix all Exact duplicates
SKY-DEMO-004	Photos do not appear on a demo	Open the page online; try another photo link in Admin site → Demo sites
SKY-DEMO-005	Fonts look plain	No action needed; fonts load when online
SKY-IMP-003	Staging table not empty	Nothing to do: each import clears staging at the start
SKY-LEAD-002	Overture extract is empty	Zoom the Overture Explorer map to the area, wait for points to appear, then run the extract again
SKY-LEAD-003	Data source blocked by the network	Use the Overture Explorer in the Claude desktop browser (works)
SKY-GIT-004	GitHub has changes this computer does not	Open GitHub Desktop → Fetch origin → Pull, then push again
SKY-GIT-005	Git is not installed or this is not the repository folder	Install Git for Windows (free) and reopen the window
SKY-BAK-002	Database damaged or lost — restore needed	Find the newest SkyTechCRM_*.bak in the backup folder

Code	Problem	Prevention already in place / check
SKY-BAK-003	No recent database backup	In SSMS run: EXEC SkyTechCRM.dbo.usp_BackupSkyTechCRM;
SKY-SYS-002	Disk full	Free disk space (empty Recycle Bin, delete old backups beyond the last 5)

6. SQL Server error numbers

The Admin site converts database errors with this table (admin-site/src/errors.js). Use it when SSMS shows a number.

SQL number / driver code	Meaning	SkyTech code
2601 / 2627	Duplicate key (unique rule)	SKY-DUP-001
547	Record is linked (foreign key)	SKY-DATA-003
50001 / 50002	usp_MergeCompanies refused	SKY-DUP-005
50011 / 50012	usp_MergeLeads refused	SKY-DUP-002
50021	usp_FixDuplicateWebPresence: nothing to fix	SKY-DUP-003
50100	AuditLog is append-only	SKY-SEC-006
208 / 2812	Object (table/view/procedure) missing	SKY-DB-005
229 / 230	Permission denied	SKY-DB-008
4060	Database not found / no access	SKY-DB-004
18456	Login failed	SKY-DB-006
8152 / 2628	Value too long	SKY-DATA-002
ETIMEOUT	Query or connection timed out	SKY-DB-007
ESOCKET / ELOGIN	Cannot reach SQL Server	SKY-DB-001
IM002	ODBC driver not found	SKY-DB-003
anything else	Unexpected database error	SKY-DB-009

7. Database error and audit queries (SSMS)

```
-- newest database errors
SELECT TOP 20 ErrorLogID, LoggedOn, ProcedureName, ErrorNumber, SkyCode, ErrorMessage, Context, LoginName
FROM SkyTechCRM.dbo.ErrorLog ORDER BY ErrorLogID DESC;

-- errors per procedure in the last 30 days
SELECT ProcedureName, SkyCode, COUNT(*) AS Errors, MAX(LoggedOn) AS Newest
FROM SkyTechCRM.dbo.ErrorLog WHERE LoggedOn > DATEADD(day, -30, GETDATE())
GROUP BY ProcedureName, SkyCode ORDER BY Errors DESC;

-- who changed a lead (audit)
SELECT TOP 50 At, Username, Action, Entity, RecordKey, Details FROM SkyTechCRM.dbo.AuditLog
WHERE Entity = N'leads' ORDER BY AuditID DESC;

-- failed sign-ins and lockouts in the last 7 days
SELECT At, Username, Action, Details FROM SkyTechCRM.dbo.AuditLog
WHERE Action IN ('LOGINFAIL', 'LOCKOUT') AND At > DATEADD(day, -7, GETDATE()) ORDER BY At DESC;

-- housekeeping (owner): keep 90 days of database errors
EXEC SkyTechCRM.dbo.usp_PurgeErrorLog @KeepDays = 90;
```

8. Manual recovery recipes

Situation	Steps
Admin site will not start	Run start-admin.bat — it runs the doctor first and stops on the first problem with its code. Typical: .env missing (SKY-CFG-001), weak SESSION_SECRET (SKY-CFG-002: npm run doctor -- --new-secret), SQL Server service stopped (SKY-DB-001: Services → SQL Server (MSSQLSERVER) → Start).
Locked out of the Admin site	Wait 15 minutes (SKY-AUTH-002), or restart the Admin site to clear the lock. Forgotten password: another admin resets it on the Logins screen; if no admin can sign in, run npm run create-admin in admin-site with the same username (this also re-enables the login).
A table or procedure is missing	In SSMS run the scripts in order: sql/01 → lead import → sql/04 → sql/05 → sql/06. All are safe to re-run and never delete data.

Situation	Steps
Duplicates appeared	Admin site → Possible duplicates: merge or dismiss each Exact pair, then re-run sql/01 so the unique rules are created (SKY-DUP-004).
Database damaged or deleted (SKY-BAK-002)	Stop the Admin site. In SSMS: RESTORE DATABASE SkyTechCRM FROM DISK = N'C:\SkyTechBackups\<newest>.bak' WITH REPLACE, CHECKSUM; then run sql/04 and sql/06 and npm run doctor.
Push to GitHub fails	Read the code printed by push-to-github.bat: lock file (SKY-GIT-002: close GitHub Desktop, delete the .lock file), secret found (SKY-SEC-004: remove it, rotate the key if it was ever pushed), sign-in (SKY-GIT-001: sign in again in GitHub Desktop), GitHub newer (SKY-GIT-004: Fetch/Pull in GitHub Desktop, then push).
A secret was uploaded by mistake	Change the password/key at its source immediately (rotation is the real fix — removing the file later does not erase GitHub history). Then remove it from the file, commit, push, and tell SkyTech_Manager to clean the history.
Disk full (SKY-SYS-002)	Delete old files in C:\SkyTechBackups (keep the newest 4) and logs\runtime older than 30 days; empty the Recycle Bin; run the doctor.
Moving to a second computer or to hosting	Copy the repository, run npm install, create a new .env (new SESSION_SECRET), keep HOST=127.0.0.1 unless the site is behind HTTPS with COOKIE_SECURE=1 and TRUST_PROXY=1, use the SkyTechApp least-privilege login (sql/06), run the doctor.

Version: V1.2 (2026-10-10) — docs/architecture/SkyTech_Troubleshooting_Guide.html — V1.2. Rebuild with: python scripts/docs/build_ops_docs.py